Anjali Roy Rathi

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Education

University of Exeter, UK — Jan 2025 – Jan 2026
MSc Statistical Data Science (Merit)

SZABIST, Pakistan — Sep 2020 – Aug 2024
BS Software Engineering, CGPA: 3.53

Skills

Languages: Python, SQL
Machine Learning: Logistic Regression, Random Forest, XGBoost, Lasso, Cross-validation
Deep Learning: CNNs, U-Net, RCAN
Data: Pandas, NumPy, Feature Engineering, EDA, Data Preprocessing
Backend: FastAPI, REST APIs, Pydantic
Tools: Git, Jupyter, VS Code

Projects

AI Resume Analyzer API (FastAPI)

- Built REST API for resume-job matching using structured inputs, scoring logic, and ATS-style keyword comparison.
- Designed Pydantic validation schemas and preprocessing pipeline for resume and job description text.
- Developed modular backend enabling scalable API integration and future LLM-based semantic matching.
- Tested endpoints using Swagger UI for real-time validation, debugging, and workflow verification.

Customer Churn Prediction System

- Developed models (Logistic Regression, Random Forest, XGBoost) on structured telecom datasets.
- Applied preprocessing, feature engineering, encoding, and missing-value handling to improve model robustness.
- Improved recall through threshold tuning for business-critical churn detection and retention decisions.
- Evaluated models using precision, recall, ROC-AUC, and confusion matrix analysis.

Fraud Detection System

- Built fraud detection models on highly imbalanced transaction datasets for rare-event prediction.
- Achieved 91% recall while reducing false positives using class weighting and threshold optimisation.
- Balanced fraud detection performance with operational efficiency using precision-recall evaluation.

MRI Super Resolution using Deep Learning

- Developed U-Net and RCAN models to enhance low-resolution MRI images from paired datasets.
- Evaluated reconstruction performance using PSNR, SSIM, and RMSE across architectures.

Statistical Modelling for Tuberculosis Risk

- Applied Generalized Additive Models on 557 datasets for interpretable disease risk analysis.
- Identified socio-economic drivers including unemployment, population density, and urbanisation.

Machine Learning Coursework

- Built regression and classification models using KNN, SVM, Decision Trees, and cross-validation.
- Applied preprocessing including encoding, normalization, feature selection, and missing-value handling.
- Evaluated model outputs using accuracy, recall, and error analysis for data-driven decision making.

Banking and ATM Simulation System

- Designed OOP-based banking workflows supporting deposits, withdrawals, balance tracking, and validation.
- Implemented error handling for invalid PIN, insufficient balance, and incorrect transaction inputs.

Certifications

Python Programming Crash Course — Sep 2023
Microsoft AI Technologies Workshop — Feb 2024